PERSONAL LAB CASE STUDY

Hybrid Identity Lab

Active Directory to Microsoft Entra ID and Microsoft 365

Harun

Lab work: 16-22 October 2026

Project summary

I connected an on-premises Active Directory lab to Microsoft Entra ID and Microsoft 365, then used the environment to work through the kinds of tasks that come up in IT support and junior systems administration: synchronization, group-based licensing, Conditional Access, password reset and writeback, sign-in troubleshooting, onboarding, offboarding, and least-privilege administration.

Lab period	16-22 October 2026
On-prem domain	lab.local (DC01, FS01, WIN11-IT-01)
Cloud tenant	jaradilab.onmicrosoft.com (Microsoft 365 Business Premium trial)
Core tools	Active Directory, Microsoft Entra Connect Sync on Windows Server 2025 (FS01), Windows 11, Microsoft Entra admin center, Microsoft 365 and Exchange admin centers

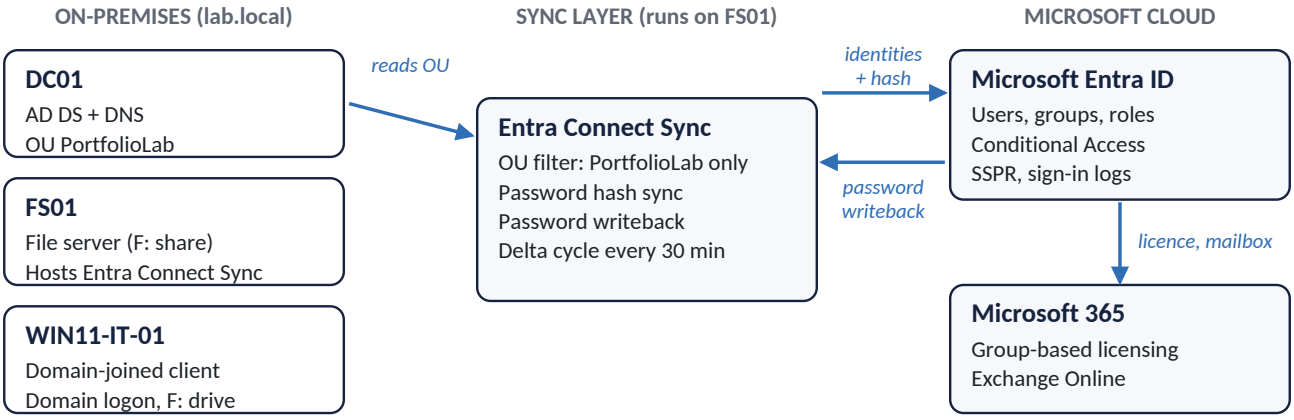
> All screenshots are from the lab. Tenant, object, session, request and correlation IDs and IP addresses are blacked out, and no passwords are shown. Every screenshot from the project is in the separate *Full Evidence Appendix*.

1. The lab at a glance

A small hybrid environment built on top of my existing Windows Server lab.

THE QUESTION

How do you connect an on-premises Active Directory domain to Microsoft 365, so each user has one identity for both?



The most important design choice was the sync scope. Only the **PortfolioLab** OU was synchronized. The on-prem admin accounts stayed outside that scope, so they never appeared in Entra ID. Synced employee accounts were still managed from Active Directory, while Entra ID and Microsoft 365 supplied the cloud services.

Identity flow

A user starts in Active Directory. Entra Connect reads the selected OU and synchronizes the identity and the password hash to Entra ID. Group membership then supplies a Microsoft 365 licence and, through it, a mailbox. For self-service password reset, password writeback sends the new password the opposite way, back to AD.

Accounts used in the lab

Account	Where it lives	Purpose
adm-ajaradi	AD only (not synced)	On-prem admin work: ADUC, Group Policy, sync commands
adm-jaradi	Cloud only	Named Global Administrator for day-to-day cloud admin
breakglass01	Cloud only	Emergency access, excluded from Conditional Access
helpdesk01	Cloud only	Helpdesk Administrator role only
finance.test01, hr.test01, s.hassan	Synced from AD	Test users for licensing, mail and helpdesk scenarios

2. Preparing Active Directory and proving synchronization

> THE QUESTION
>
> **Is the AD data ready to sync, does the sync really work, and which directory owns a synced user's details?**

Before the first sync I checked the directory instead of assuming the data was ready. Using PowerShell, I found three accounts in PortfolioLab with no UserPrincipalName, and corrected them in ADUC before synchronizing. The original workbook suggested IdFix, but I audited the required AD attributes directly with PowerShell and ADUC instead. I also kept the lab's *.local* domain instead of pretending it was a production domain, so synced users sign in to Microsoft 365 with the tenant's *onmicrosoft.com* name.

- Entra Connect Sync was installed on FS01, not on the domain controller.
- The connector was filtered to the PortfolioLab OU only.
- Password hash sync and password writeback were enabled.
- I checked the sync scheduler (30-minute delta cycle) and ran a manual delta sync when I needed to test straight away.
- I changed a synced user's job title in AD and then saw it in Entra as read-only cloud data. That confirmed Active Directory is the source of authority for synced users.

> **A real problem during setup**
>
> The Entra Connect wizard failed at the sign-in step with WAM error 3399614467. Starting the wizard with the */InteractiveAuth* switch let the authentication step complete. I kept this in the case study because the troubleshooting was more useful than a perfectly clean setup.

```
PS C:\WINDOWS\system32> # Load Active Directory commands
Import-Module ActiveDirectory

# Find users inside PortfolioLab, including Finance and HR
Get-ADUser -Filter * `
    -SearchBase "OU=PortfolioLab,DC=lab,DC=local" `
    -Properties UserPrincipalName,Mail |
    Select-Object Name,SamAccountName,Enabled,UserPrincipalName,Mail


Name              : Finance Reviewer
SamAccountName    : finance.reviewer
Enabled           : True
UserPrincipalName : finance.reviewer@lab.local
Mail              :

Name              : Finance Test01
SamAccountName    : finance.test01
Enabled           : True
UserPrincipalName :
Mail              :

Name              : HR Test01
SamAccountName    : hr.test01
Enabled           : True
UserPrincipalName :
Mail              :

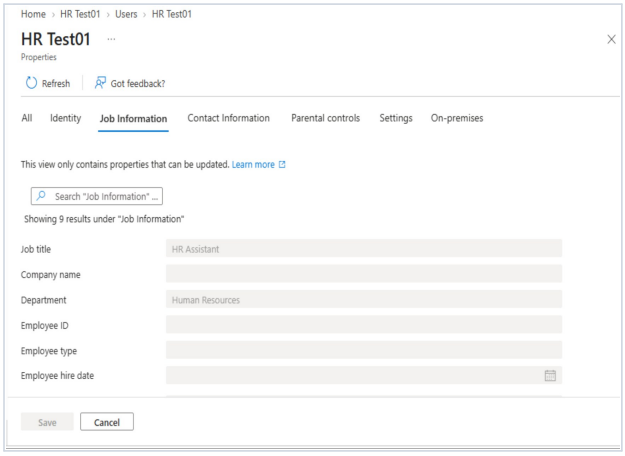
Name              : Jonas Helpdesk
SamAccountName    : jonas
Enabled           : True
UserPrincipalName :
Mail              :

Name              : NoAccess Test01
SamAccountName    : noaccess.test01
Enabled           : True
UserPrincipalName : noaccess.test01@lab.local
Mail              :



PS C:\WINDOWS\system32>
```

Before the first sync: three users had an empty UserPrincipalName.



A job title changed in AD shows in Entra as read-only: AD owns it.

Why this matters: in a hybrid incident, the first questions are usually whether the object is in sync scope, whether synchronization is healthy, and which directory owns the attribute that needs to change.

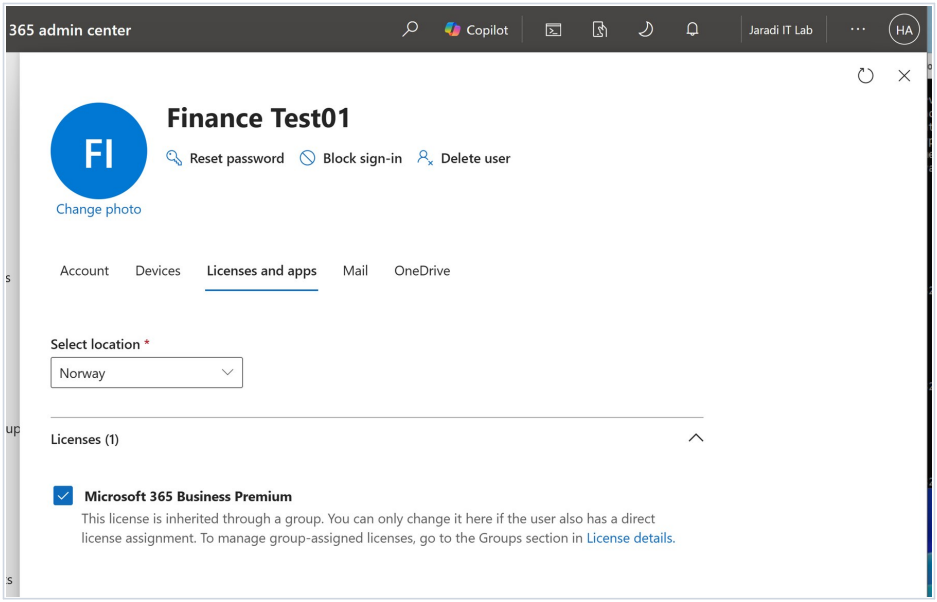
3. Licensing and mail through groups

THE QUESTION

How do you give every user a Microsoft 365 licence and mailbox without assigning them one by one?

Licensing was group-based rather than assigned user by user. I created the AD security group **GG_M365_BusinessPremium**, added the test users as direct members, synchronized the group, and assigned Microsoft 365 Business Premium to the cloud copy of that group. The users inherited the licence automatically and Exchange Online created their mailboxes.

- The Finance and HR test users received Business Premium through group membership.
- The licence page showed no licensing errors and no members without licences.
- The users' usage location showed Norway, which a licence needs before it can be assigned.
- Mail was tested in both directions between the two accounts in Outlook on the web.
- The same model was later reused for the new starter and the leaver.



The licence is inherited through the group, not assigned directly.

What I took from this

For routine onboarding, the important thing is not remembering where the licence button is. It is understanding the chain: AD group membership, then synchronization, then the cloud licence, then the mailbox.

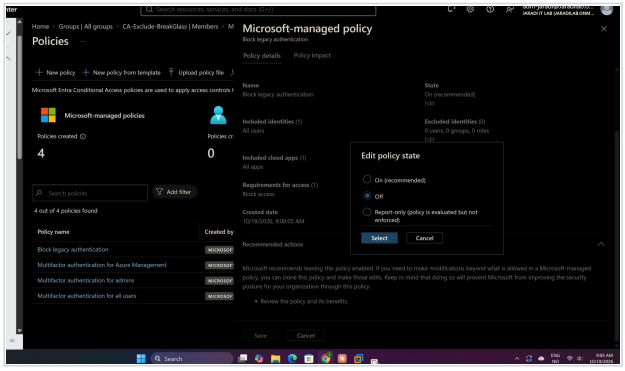
4. Conditional Access without locking out the tenant

THE QUESTION

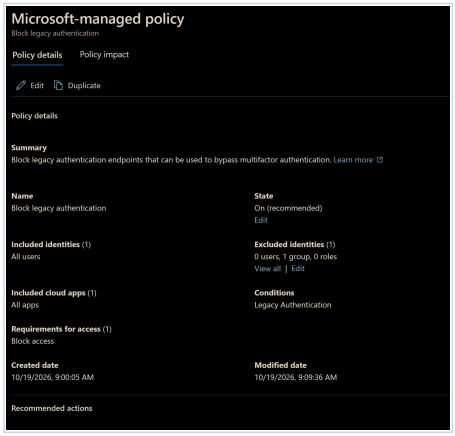
How do you require MFA and block legacy sign-in without locking yourself out of the tenant?

I replaced Security Defaults with my own Conditional Access policies, but tested them before enforcing them. **CA001** requires MFA for all users, excluding the break-glass group and the Directory Synchronization Accounts role. **CA002** blocks legacy authentication by targeting only Exchange ActiveSync and "other" clients. Both started in report-only mode so I could read the sign-in results before switching them on.

One unexpected detail became an important part of the lab. When I disabled Security Defaults using the replacement option, four Microsoft-managed policies appeared in the tenant. They were already On, and they did not exclude the break-glass account. Instead of turning them off and leaving the tenant without MFA, I added the emergency-access group to each policy's exclusions. The managed policies stayed On, alongside CA001 and CA002, for the rest of the lab.



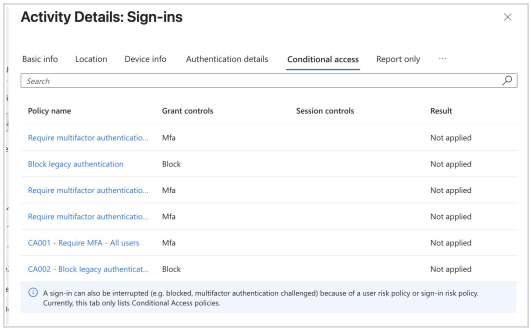
As it first appeared: no excluded identities. (The Off dialog shown was cancelled.)



After the fix: break-glass group excluded, policy still On.

How I verified it

A normal test user's sign-in log showed CA001 applied and satisfied with MFA. A separate sign-in with the break-glass account showed the Conditional Access policies as not applied. The emergency account can still be asked for MFA by Microsoft's own admin-portal requirement, which is separate from my policies.



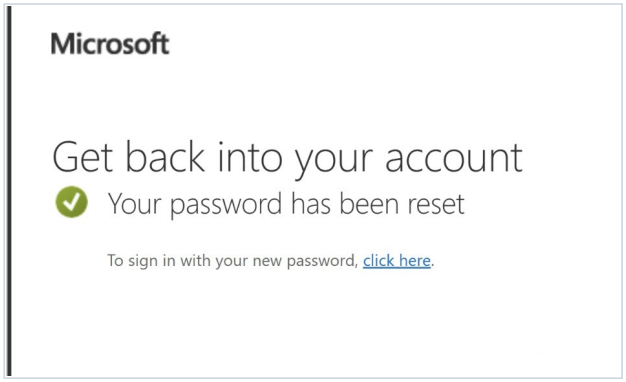
Break-glass sign-in: CA001, CA002 and the managed policies are not applied.

5. Self-service password reset and writeback

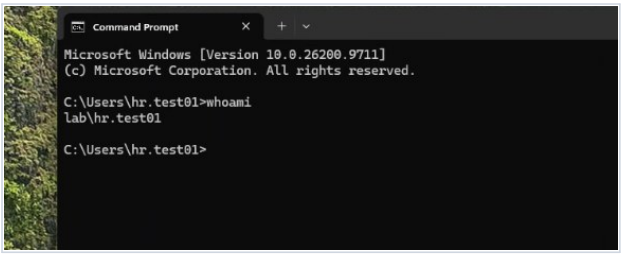
THE QUESTION

Can users reset a forgotten password themselves, and will the new password also work on their domain PC?

The SSPR test was meant to prove more than a cloud password change. SSPR was enabled for the Microsoft 365 licence group only, password writeback was enabled, and a normal test user registered and then reset the password through Microsoft's self-service page. The real proof came afterwards: the new password worked for the domain logon on WIN11-IT-01.

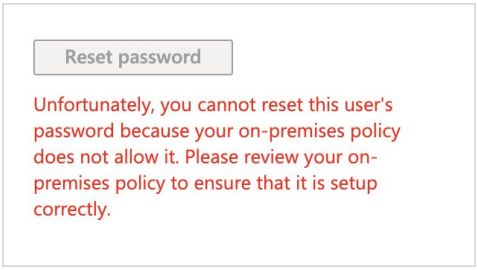


Cloud self-service reset completed.



The same account signed in on the domain PC with the new password.

The domain policy mattered during later reset tests. When an admin reset from the Entra admin center failed with "on-premises policy does not allow it", the message did not name the rule that failed. A longer temporary password was later accepted, which showed that the writeback path was working and that AD was enforcing its own password policy.



The Entra admin reset rejected by on-prem AD policy.

Lab policy note

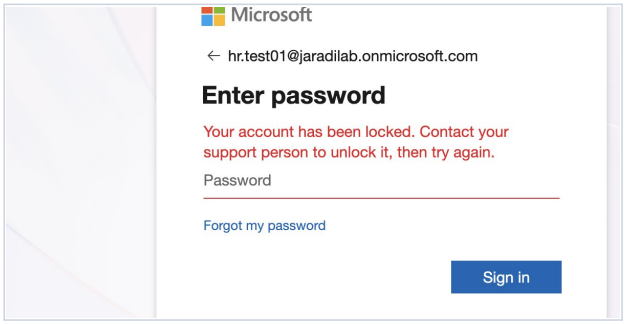
Minimum password age stayed at 1 day and the default minimum length was 7 characters. Microsoft recommends a minimum age of 0 for writeback, because otherwise a user cannot reset twice in 24 hours. I kept the existing domain-wide policy and recorded the trade-off instead of weakening it to make a test easier.

6. Troubleshooting sign-in problems from the logs

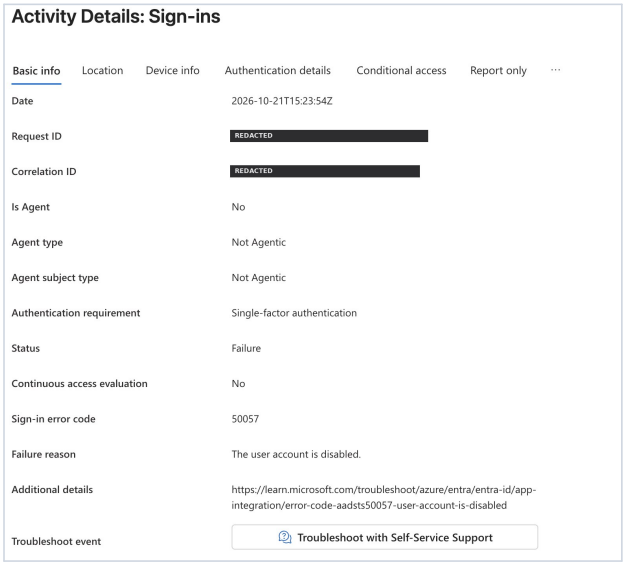
THE QUESTION

A user says "I can't sign in". How do you find the real cause before changing anything?

The most useful helpdesk lesson in the project was to read the sign-in log before changing the account. In one test the sign-in page told the user the account was "locked". The Entra sign-in log showed error **50057** instead: the account was disabled. The right fix was to enable it in AD and sync the change, not to look for an AD lockout that did not exist.



What the user saw: "account locked".



What the log showed: error 50057, the account is disabled.

A second test used three wrong passwords. Those attempts appeared as error **50126**, invalid username or password, and the correct password afterwards still worked. Comparing the two cases made the value of the log clear: similar user complaints can have completely different causes.

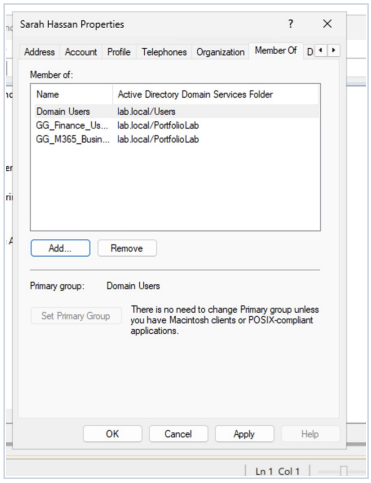
Support habit I want to keep

Start with the evidence: timestamp, user, application, status, error code and Conditional Access result. Change the account only after the failure has been identified.

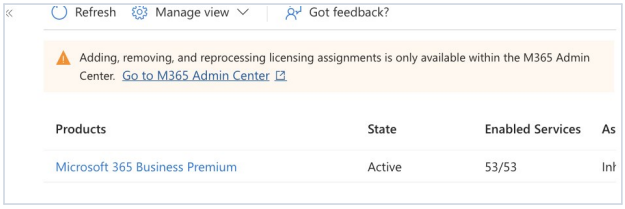
7. Onboarding a new starter

> **THE QUESTION**
>
> **A new Finance employee starts on Monday and needs a PC logon, the F: drive and email. What has to happen?**

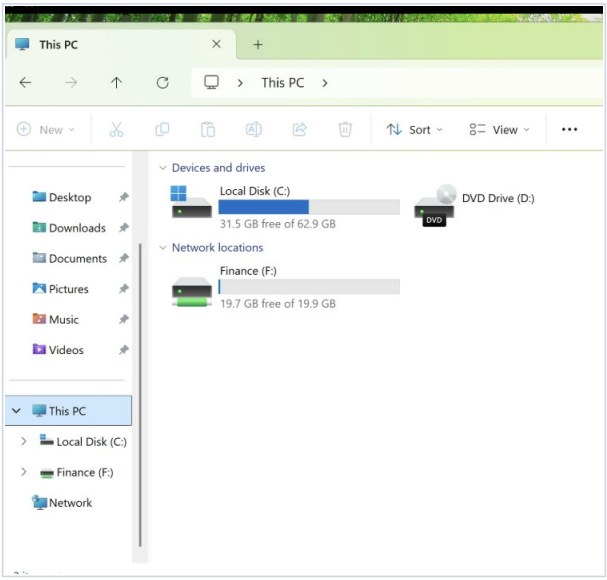
For the new-starter test I created **Sarah Hassan** in the Finance OU and used group membership to give both file access and Microsoft 365: *GG_Finance_Users* for the Finance share and *GG_M365_BusinessPremium* for the licence. After a delta sync the account was licensed automatically, a mailbox was created, MFA was registered, and the Finance drive was available on the domain PC.



The new starter's AD groups: file share and licence.



Business Premium inherited after the sync.



Finance (F:) share mapped on WIN11-IT-01 during the new-starter test.

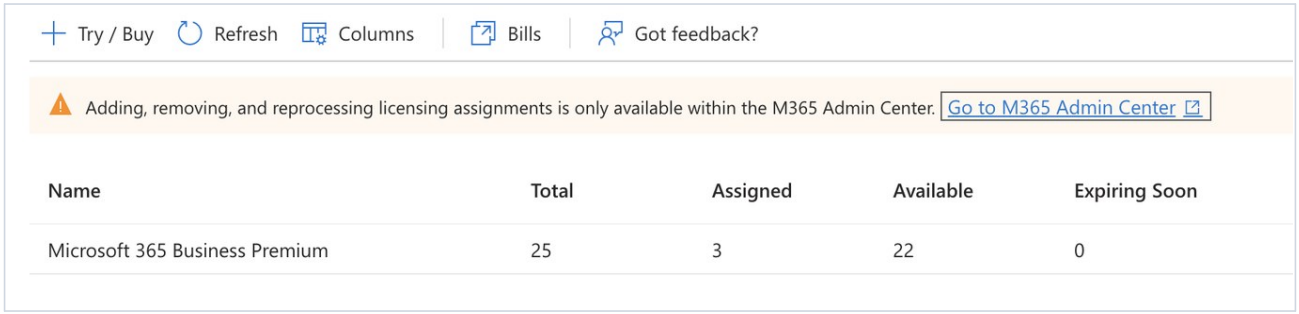
8. Offboarding the same account

The same identity was used to test a controlled leaver process.

> THE QUESTION
>
> **An employee leaves today. Their manager needs their email, the licence must be freed, and the account must stop working. In what order do you do it?**

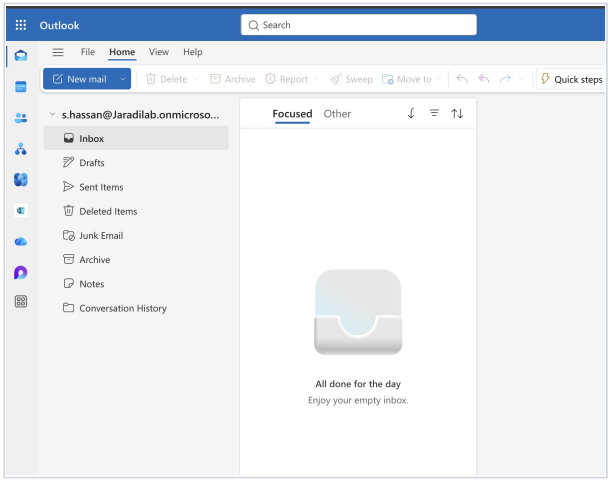
The order mattered, so I followed it step by step:

- Converted the mailbox to a shared mailbox **while it was still licensed** (Microsoft requires the licence for the conversion).
- Gave the manager, Finance Test01, Full Access to it.
- Revoked all active sessions.
- Disabled the account in AD and left it in the synced OU, because the cloud identity anchors the shared mailbox.
- Removed the user from the licence group and ran a delta sync.
- Checked the result: sign-in blocked, the manager could open the mailbox, and Business Premium usage dropped from 4 to 3 of 25.

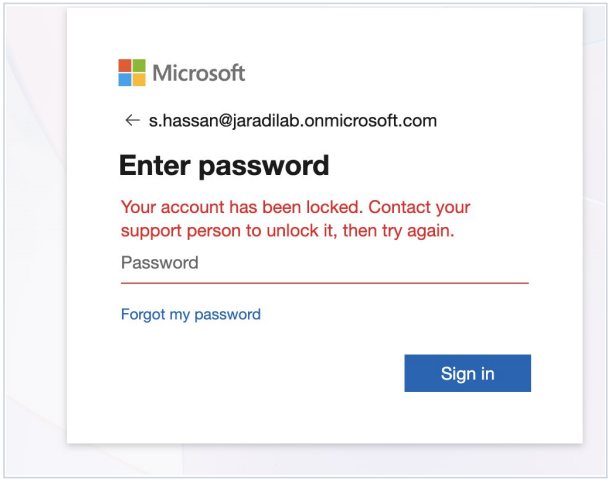


Name	Total	Assigned	Available	Expiring Soon
Microsoft 365 Business Premium	25	3	22	0

Business Premium after the leaver's licence was freed: 25 total, 3 assigned, 22 available.



Finance Test01 opening the former user's shared mailbox.



The leaver can no longer sign in.

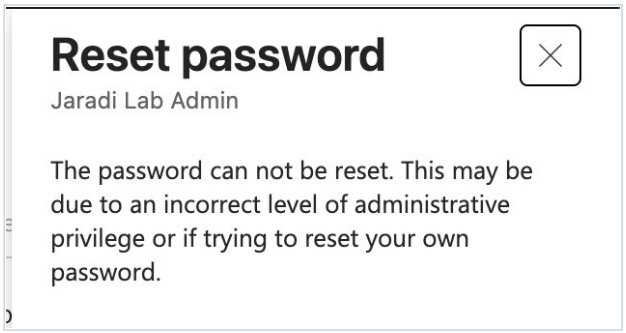
9. Least privilege for the helpdesk

THE QUESTION

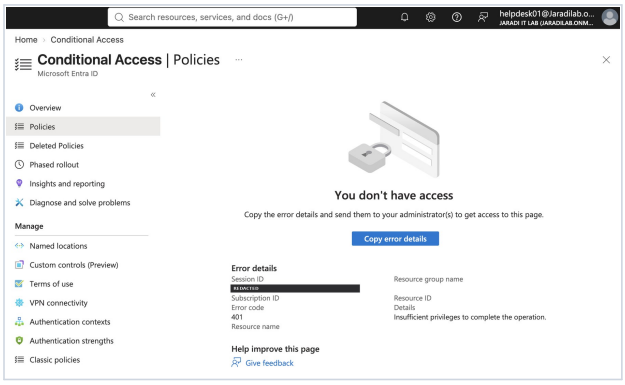
How do you let a helpdesk colleague reset passwords without giving them full admin rights?

The last admin test used a separate cloud-only account, **helpdesk01**, with the Helpdesk Administrator role only. The point was to find the edge of the role, not just to confirm that an admin account can do everything.

- It reset a normal user's password successfully.
- It could not reset the named Global Administrator's password.
- It could open the Conditional Access area, but the policies page returned 401 / insufficient privileges.



Password reset blocked for the Global Administrator.



Conditional Access blocked: insufficient privileges.

How the three roles compare

Role	Reset a normal user	Reset a Global Admin	Manage Conditional Access	Source
Helpdesk Administrator	Yes	No	No	Tested in this lab
User Administrator	Yes	No (non-admins and limited admins only)	No	Microsoft Learn, not tested
Global Administrator	Yes	Yes	Yes	Microsoft Learn; used throughout the lab

That was the result I wanted from the role design: enough authority to solve a normal password ticket, but not enough to change high-privilege accounts or the tenant's sign-in policy.

10. What I would change in production

In this lab	In a production environment
One cloud-only emergency account	At least two emergency accounts, protected with phishing-resistant authentication and monitored with sign-in alerts
lab.local plus onmicrosoft.com sign-in names	A verified public domain and matching UPN suffixes
Permanent Global Administrator assignments	Just-in-time privileged roles where licensing and design allow it
Default 7-character minimum password length	A stronger password policy aligned with the organization's security standard
AD Recycle Bin disabled	Enable it and test recovery before production use
Entra Connect Sync on a single lab VM	Documented resilience, monitoring, backup and change control; also evaluate Microsoft Entra Cloud Sync, which Microsoft suggests considering as the successor

> **What changed for me after this lab**
> I am more comfortable treating AD and Entra ID as one identity system with two control planes. I also know better where a helpdesk technician should look first: AD for the source account, Entra for sign-in evidence and roles, Microsoft 365 for licensing and mail, and the sync server when those two worlds do not agree.

Skills used

Active Directory administration, Microsoft Entra Connect Sync, Microsoft Entra ID, Conditional Access, MFA, SSPR and password writeback, Microsoft 365 group-based licensing, Exchange Online mailboxes and delegation, sign-in log troubleshooting, onboarding and offboarding, and least-privilege role assignment.

11. Scope notes

Two points about what the tests did and did not cover.

Admin reset of the first test user. After AD rejected the temporary password in the Entra admin reset for Finance Test01, I did not repeat that exact reset. A later helpdesk password reset succeeded with a longer, policy-compliant temporary password.

MFA reset audit trail. The audit log evidence shows the user's MFA re-registration. The admin-side audit entries for the re-register and revoke-sessions actions are not part of the evidence shown here.